

A UNIVERSAL ALGORITHM FOR CALCULATION OF NUTRIENT SOLUTIONS

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Abstract

An algorithm for the calculation of nutrient solutions is presented based on standard nutrient solutions as described elsewhere by the Research Station at Naaldwijk. These nutrient solutions are based on the average uptake of minerals of specific crops. The algorithm shows possibilities for adjustments to special requirements of the crop like differences in the uptake depending on growth stage, crop development and growing conditions. It also supplies possibilities for adjustment of the mutual ratios of nutrients if the total concentration of nutrients (EC-value) is increased. If no adjustments are made in such cases, the mutual ratios between the anions and the cations are kept constant, except for phosphorus and ammonium. The concentration of these ions are kept constant with increasing EC-values. This is also the case with the micro nutrients. The algorithm is suitable for different fertilizer combinations and shows excellent possibilities for adjustment to the chemical composition of the raw water. It adjusts the application of acids in relation to the water quality and the EC-value in the nutrient solution entirely given to the plant.

The paper also gives information about adjustments necessary during the growing period of the plant. To this purpose a regular sampling and analysis of the nutrient solution in the root environment is desirable. Partly these analyses can be carried out by the grower himself with portable instruments, like the determination of pH- and EC-value. However, the determination of nutrient elements generally has to be carried out under laboratory conditions. In situ measurements or measurements with portable instruments to this purpose are disappointing up till now.

1. Introduction

This paper presents an algorithm for the calculation of nutrient solutions for soilless cultures. The algorithm described was developed at the Research Station for Floriculture and Glasshouse Vegetables at Naaldwijk, The Netherlands, and is based on the results of many experiments and experience with the use of nutrient solutions on commercial nurseries. In some earlier publications (Sonneveld 1989; Sonneveld and Straver 1994) a system to calculate nutrient solutions has already been described. However, with this system it was only possible to calculate just one solution for a crop. The present system is more flexible and shows excellent possibilities for the calculation of nutrient solutions better tuned to the growing conditions and the adjustments to special requirements of the crops during the growing period.

2. Characteristics used

The algorithm is set up to reach fixed (guide) values in the root environment. To this purpose following characteristics have been developed.

2.1. Standard nutrient solutions

Standard nutrient solutions for different crops have been published by the Research Station for Floriculture and Glasshouse Vegetables by Sonneveld and Straver (1994). The nutrient solutions are characterized by the EC-value, given as dS m^{-1} at 25°C , the macro elements expressed as mmol l^{-1} and the micro elements expressed as $\mu\text{mol l}^{-1}$. The sum of the valencies of the cations will be denoted as C^+ and the sum of the valencies of the anions as A^- ; both are expressed as mmol l^{-1} . They are calculated as follows $\text{C}^+ = \Sigma (\text{NH}_4 + \text{K} + 2\text{Ca} + 2\text{Mg})$ and $\text{A}^- = \Sigma (\text{NO}_3 + 2\text{SO}_4 + \text{H}_2\text{PO}_4)$, respectively. A nutrient solution can be calculated only if $\text{C}^+ = \text{A}^-$. Furthermore the standard nutrient solutions are distinguished for crops and sometimes for the growth system used, for example a free drainage system and a recirculating system need different nutrient solutions to reach the same guide values in the root environment.

2.2. Adjustments of nutrients

The compositions of the standard nutrient solutions are tuned to meet the target values for the concentration of the different nutrients in the root environment. However, depending on the growth stage, growing conditions or results of analytical data of a routine test of the solution in the root environment, adjustments may be necessary. Such adjustments are expressed as mmol l^{-1} and mostly do not exceed 30% of the values in the standard nutrient solution for ions with a high absorption rate, like NO_3 , K and P. For Ca, Mg and SO_4 adjustments of 50% are given (Anonymus 1994).

2.3. Adjustment of EC

Depending on light intensity and nutrient uptake the EC value of the nutrient solution will fluctuate in order to reach the guide values. This means that also the addition of free acids will change if the recipe calculated contains such compounds. This can be harmful to plants when the pH in the nutrient solution drops too much and irrigation systems are blocked quickly if the pH in the solution is too high. In the algorithm the acid dosage is adjusted in such a way, that it is independent of the EC value of the nutrient solution added to the plant. Furthermore it increases or decreases the EC value by a proportional increase or decrease, respectively, of all anions and cations except H_2PO_4 and NH_4 . The addition of micronutrients is also kept constant with fluctuating EC value.

2.4. Calculation of the EC

For the calculations made in the algorithm it is sometimes necessary to estimate the EC value from the ion composition, or alternatively to calculate the sum of ions from the EC value. Between C^+ or A^- on the one hand and the EC value on the other hand there is a close correlation. Sonneveld and Van der Wees (1990) published the following relationship for C^+ .

$$\text{EC}_c = 0.095 \text{C}^+ + 0.19 \quad r = 0.96 \quad (1)$$

In which EC_c is the EC value calculated in dS m^{-1} and C^+ the sum of cations as defined under 1.1, expressed in mmol l^{-1} .

For current use the formula is simplified to

$$\text{EC}_c = 0.1 \text{C}^+ \quad (1a)$$

which is acceptable in the usual range of 1.5 to 4.0 dS m^{-1} . In case of exceptional EC values or special ionic compositions it is better to use the calculations of McNeal *et al.* (1970). Furthermore, to simplify the calculations, the contribution of a certain ion is

calculated to be equal and linear with the concentration and the valency. So from formula (1) it can be calculated that the valency of 1 mmol l^{-1} of any anion going together with the valency of 1 mmol l^{-1} of any cation is equivalent to about 0.1 dS m^{-1} , which is not exact, but the errors have no real consequences for the overall results of the calculations.

2.5. Adjustment of the pH

The pH of the nutrient solution supplied to a system should not be higher than 6.0 and not lower than 5.0. Low pH values in the root environment are harmful to the roots and high pH values in the nutrient solution cause precipitation of some nutrients like phosphates, calcium and iron. The pH in the root environment is mainly adjusted by the quantity of NH_4 added. The supply of NH_4 is mostly not harmful for crops if it does not exceed 10 - 15 % of the total nitrogen supplied. HCO_3 available in the water is neutralized with H_3O , see under "water quality". If use is made of a fertilizer dilution system with a "residual" acid requirement to tune the pH value exact to the value aimed at, no H_3PO_4 will be used, to prevent an overdosing of P.

2.6. Water quality

Adjustments with respect to water quality are made if the raw water contains nutrient elements. These nutrient elements are deducted from the nutrients that should be supplied in the nutrient solution added to the plant. Bicarbonate (HCO_3) in the water should be replaced by acids (H_3O) and taken into account with the calculations. The C^+ and the A^- of the corrections from the raw water should be equal otherwise the nutrient solution cannot be calculated.

3. Description of the algorithm

3.1. Definitions

The following definitions are used.

A	Sum of anions in mol
C^+	Sum of cations in mol
EC	EC value of a solution, expressed as dS m^{-1}
EC_c	EC value of a solution calculated from C^+ or A^- by formula (1a), expressed as dS m^{-1}
EC_{irr}	The EC value wanted in the irrigation water, expressed as dS m^{-1}
$\text{C}(\text{X})$	The concentration of ion X in the standard nutrient solution, expressed as mmol l^{-1}
$\text{C}'(\text{X})$	The concentration of ion X after the adjustment calculations keeping the C^+ and the A^- constant
$\text{C}''(\text{X})$	The concentration of ion X after adjustment on EC_{irr}
$\text{V}(\text{X})$	The valency of ion X
C_a	The concentration adjusted for NH_4 or H_2PO_4
C_b	The concentration adjusted for any ion other than NH_4 and H_2PO_4

3.2. Calculations

The following calculations are presented in the algorithm. The steps are carried out in the order mentioned.

- Selection of the standard nutrient solution, depending on crop and growing system.
- Adjustment of the mutual ratios of the macro elements of the standard nutrient solution on corrections necessary, keeping C^+ and A^- constant.
- Adjustments of the micronutrients.

- Adjustment of the EC value if the value wanted in the irrigation water at the moment differs from this given with the standard solution.
- Adjustment to the water quality.

3.2.1. Selection of the standard nutrient solution

The standard nutrient solution is selected on the basis of the crop and the growing system (Sonneveld and Straver 1994). For the growing system only two distinctions are generally made: a free drainage system and a recirculation system.

3.2.2. Adjustment of the mutual ratios of the macro elements

With adjustments on the standard nutrient solution A^- and C^+ are kept constant, as is the case with NH_4 and H_2PO_4 . So the increase or decrease of A^- and C^+ caused by the adjustments is equally settled over the other anions or cations, respectively.

An adjustment of a NH_4 with $C_a \text{ mmol l}^{-1}$ and an adjustment of another cation (X) with $C_b \text{ mmol l}^{-1}$ induces following calculations to keep C^+ of the nutrient solution equal to those of the standard nutrient solution. All concentrations except NH_4 should be recalculated as follows:

$$C'(X) = \{C(X) + C_b(X)\} * \{C^+ - C(NH_4) - C_a(NH_4)\} / \{C^+ - C(NH_4) + \sum V(X) * C_b(X)\} \quad (2)$$

The same calculation will be made in case that there are adjustments with one or more anions. An adjustment of H_2PO_4 with $C_a \text{ mmol l}^{-1}$ and another anion (X) with $C_b \text{ mmol l}^{-1}$ induces following calculation to keep A^- constant.

$$C'(X) = \{C(X) + C_b(X)\} * \{A^- - C(H_2PO_4) - C_a(H_2PO_4)\} / \{A^- - C(H_2PO_4) + \sum V(X) * C_b(X)\} \quad (3)$$

In this formula H_2PO_4 has the same position as NH_4 in formula (2).

Furthermore it will be clear that $C'(NH_4) = C(NH_4) + C_a(NH_4)$ and that $C'(H_2PO_4) = C(H_2PO_4) + C_a(H_2PO_4)$.

3.2.3. Adjustment of the micro elements

The adjustments of the micro elements on the standard nutrient solution can be made by a simple addition or deduction. The quantities of the accompanying ions can be ignored.

3.2.4. Adjustment of the EC value

In case that the EC value wanted in the irrigation solution differs from those in the standard solution, all cation concentrations except $C'(NH_4)$ should be recalculated as follows:

$$C''(X) = \{10 * C_{irr} - C'(NH_4)\} / \{10 * EC_c - C'(NH_4)\} \quad (4)$$

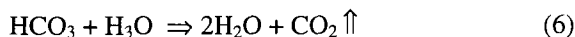
and all anion concentrations except $C'(H_2PO_4)$ should be multiplied with

$$C''(X) = \{10 * EC_{irr} - C'(H_2PO_4)\} / \{10 * EC_c - C'(H_2PO_4)\} \quad (5)$$

3.2.5. Adjustment to the water quality

After the adjustments are made, the nutrient solution obtained should be corrected for components available in the raw water. This is done by subtraction of the concentrations

found in the raw water from those calculated in the nutrient solution after adjustment of the EC value on EC_{irr} . Furthermore a correction should be made for the HCO_3 present in the raw water by addition of equivalent quantities of acid. With the addition of acid the following reaction will take place.



3.2.6. Fertilizer calculations

The calculations of the fertilizer composition of a nutrient solution from the ionic composition as given in Appendix 1 are described by Sonneveld (1989).

4. Assumptions behind the algorithm

4.1. EC value

The EC value given with the standard nutrient solution is an average value necessary to keep the EC value in the root environment on the guide value level during the growing period. Depending on the growth stage and the growing conditions adjustments will be necessary. This for example is the case with the high EC values maintained in the root environment for vegetable fruit crops under poor light conditions, to improve fruit setting or to improve fruit quality (Sonneveld 1995). In periods of high transpiration of crops mostly the concentration in the irrigation water can be a little lower than the value recommended in the standard solution.

4.2. pH value

For most crops a pH value in the root environment between 5 and 6 seems to be optimal (Voogt 1993; Voogt and Sonneveld 1997). If all the N in the nutrient solution is given as NO_3 most crops show an excessive anion demand, which induces an increase in the pH value. For that reason all standard nutrient solutions contain part of the N in the form of NH_4 . The quantity of NH_4 necessary to keep the pH under control varies e.g. with the growth stage of the crop and the weather conditions. With low pH values root damage can occur and high pH values induce chlorosis and yield reduction. Control of the pH value in the root environment is only possible for a small part by the adjustment of the pH of the nutrient solution added, since the pH value of this solution should be kept between 5.0 and 6.2. This means that the quantities of free acid that can be added in this way are restricted. For the greater part the adjustment of the pH should be achieved by the regulation of NH_4 addition. If the pH values in the root environment become too high extra NH_4 is given to aggravate the cation uptake by the crop and with too low pH values in the root environment the quantity is diminished. The NH_4 level used in the standard nutrient solutions varies for most crops between 7 and 14% of the total N supply. During the growing period the supply will vary between 0 and 20%, due to the course of the pH value in the root environment. With crops sensitive to Ca deficiency it is wise to be careful with the addition of NH_4 . In such cases it is better to accept a little higher pH in the root environment than undertake a risk on Ca deficiency.

4.3. K, Ca and Mg ratio

Standard nutrient solutions are tuned to the average uptake of nutrients of the crop in question. Incidentally crops will show high absorptions of certain elements. This is for example the case with fruit bearing crops in periods of heavy fruit load and with flower crops during strong stalk development. For some crops models have been developed to meet expected high or low absorptions (Voogt 1993a).

4.4. Anions

With respect to the anions it is important to prevent that the P concentration in the root environment becomes too high. High phosphate content can induce a too high absorption of this element by some plants and seems to be toxic (Howell and Bernhard 1961; Zijlstra *et al.* 1987). Furthermore it is found that high phosphate induces gold specks in vegetable fruit crops, which has a very negative effect on fruit quality (De Kreij *et al.* 1992). For this reason P is excluded from the adjustments calculated by in the formulae (3) and (5). Many crops absorb only little SO_4 , which results in accumulation of this ion in the root environment, especially in case of recirculation. This is stimulated by the fact that SO_4 is not harmful to crops and so residual differences in the ionic balans of nutrient solutions are balanced by the addition of some extra SO_4 . So, often reductions are necessary, especially in recirculating systems.

4.5. Micro nutrients

The addition of the micro nutrients is kept independent from the addition of the macro nutrients. This is done because the uptake of these elements is not strongly affected by changes of the concentrations of the macro nutrients. Moreover, B and especially Mn can soon achieve a level toxic to crops sensitive to high contents of these elements in the root environment. Beside differences between crops, mostly also great differences between cultivars occur. Iron should be added as a chelate. In The Netherlands DTPA is used. EDTA is also usable, but Fe chelated in this form at higher pH values is a little less stable than chelated with DTPA (Lindsay *et al.* 1967). In some cases when pH values above 6.5 are noticed in the root environment the use of EDDHA will be considered.

4.6. Root environment

The composition of the nutrient solution in the root environment is an important factor to control the nutrient supply to the crops. Comparison of the composition of the standard nutrient solution with the guide values of the solution in the root environment shows that the ratios between the values differ strongly (Sonneveld and Straver 1994). The differences between the composition of the standard nutrient solution and the guide values in the root environment are caused by the fact that there is a great difference between the rate of uptake between ions. With macro elements for example the monovalent ions are absorbed much easier than the divalent ones. Therefore the concentration of the divalent ions should be relatively high in the root environment to accelerate the uptake. Comparable situations occur with the ratios between micro nutrients. Mn is absorbed easily, while for most crops the uptake rate of B is low. So, Mn is relatively low and B is relatively high in the root environment in comparison with the concentrations in the standard solution. However, such situations can differ for crops; for rose, for example, high B contents in the root environment easily induce toxicity (De Kreij *et al.* 1985). The figures listed in Table 1 clearly show the differences existing between crops and nutrient solutions.

The composition of the solution in the root environment should be carefully taken into account for the adjustments of the addition of the nutrients to the irrigation water. To this purpose the grower should measure the EC and pH values of the solution in the root environment himself two or three times a week. Moreover, every fortnight a sample should be analysed for nutrient elements. Comparison of these data with the guide values offers valuable information for the adjustments necessary.

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Table 1. The composition of the standard nutrient solutions for rose and tomato grown in recirculation systems in comparison with guide values recommended for the nutrient solution in the root environment.

Elements	Tomato		Rose	
	St NS*	GV RE**	St NS*	GV RE**
EC	1.6	4.0	0.7	2.2
NH ₄	1.0	<0.5	0.85	<0.5
K	6.5	8.0	2.15	5.0
Ca	2.75	10.0	0.9	5.0
Mg	1.0	4.5	0.5	3.0
NO ₃	10.75	23.0	4.3	12.5
H ₂ PO ₄	1.25	1.0	0.5	0.9
SO ₄	1.5	6.75	0.5	3.0
Fe	15	25	15	25
Mn	10	5	5	3
Zn	4	7	3	3.5
B	20	50	15	20
Cu	0.75	0.7	0.5	1.0
Mo	0.5	-	0.5	-

* Standard nutrient solution

** Guide values in the root environment

Appendix 1

An example of the calculation of a nutrient solution

Characteristics

Standard nutrient solution for tomatoes in a recirculation system

Extra addition of 2 mmol l⁻¹ K and of 10 μmol l⁻¹ B is necessary

EC_{irr} wanted is 1.8

For adjustment of the pH in the root environment an extra application of 0.5 mmol NH₄ is necessary

The raw water contains the following components in mmol l⁻¹: 1.0 Ca, 0.4 Mg, 0.3 SO₄ and 2.2 HCO₃.

Calculations

Elements	Columns*					
	A	B	C	D	E	F
EC	1.6		1.6	1.8		1.8
NH ₄	1.0	+0.5	1.5	1.5		1.5
K	6.5	+2.0	7.17	8.76		8.76
Ca	2.75		2.32	2.84	1.0	1.84
Mg	1.0		0.84	1.03	0.4	0.63
NO ₃	10.75		10.75	13.10		13.10
H ₂ PO ₄	1.25		1.25	1.25		1.25
SO ₄	1.5		1.5	1.83	0.3	1.53
HCO ₃					2.2	
H ₃ O						2.2
Fe	15		15	15		15
Mn	10		10	10		10
Zn	4		4	4		4
B	20	+10	30	30		30
Cu	0.75		0.75	0.75		0.75
Mo	0.5		0.5	0.5		0.5

*Steps in the calculations. A - the standard nutrient solution (Sonneveld and Straver 1994), B - corrections necessary in the nutrient solution, C - results of the calculations made with respect to the corrections (formulae 2 and 3), D - results of the calculations made with respect to the adjustment to the EC_{irr} wanted (formulae 4 and 5), E - composition of the raw water, F - entire composition of nutrient solution to calculate.